

LESSON 8.11

SURFACE AREA OF A RECTANGULAR PYRAMID



LESSON INTRODUCTION

MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.

LEARNING TARGETS

- I can find the surface area of a rectangular pyramid using a net or the figure.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

ESSENTIAL QUESTIONS

- What are the two-dimensional shapes that create the net of a rectangular pyramid?
- What is the process for finding surface area of a pyramid?
- How can the area of each face on a net be used to find the surface area of a rectangular pyramid?

LESSON NARRATIVE

This lesson begins with an activity justifying a bedroom preference based on mathematical characteristics like perimeter and area. Students connect prior knowledge of surface area to rectangular pyramids through the real-world example of the Egyptian pyramids. Students practice drawing nets of various pyramids and are guided through finding the surface area of pyramids both as solids and as nets. Students synthesize their learning through 8.11.3 Collaboration and individually through the 8.11.5 Wrap-Up finding surface area of a pyramid.

COMPONENT SUMMARY

8.11.1 WARM-UP (~5 MIN.)

Would You Rather activity

8.11.3 COLLABORATION (~5 MIN.)

Collaborate to find the surface area of a pyramid

8.11.5 WRAP-UP (~5 MIN.)

Individual assessment on finding surface area of a pyramid

8.11.2 GUIDED INSTRUCTION (15-20 MIN.)

Introduction to pyramids and finding the surface area using nets

8.11.4 YOUR TURN! (10-15 MIN.)

Practice finding the surface area of three pyramids

RESOURCES

GLOSSARY

Key Terms:

- Slant height

Additional Terms:

- Surface area
- Face

MATERIALS

- Four-function calculator
- (Optional) Graph paper to draw and cut out nets or preprinted nets

ADDITIONAL PREPARATION

- Examples of nets of square pyramids and of rectangular pyramids may be beneficial.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse the height and the slant height.
- Students may assume that all the triangles have the same dimensions or the same area.
- Students might not remember that all sides of a square are equal in length.

THINGS TO CONSIDER

- Students may only make selections based on preferences rather than on the math provided during the 8.11.1 Warm-Up. Try not to force students to use mathematics to determine their answer. Allow students to enter the problem and choose their own method of selection.
- Students who are struggling may need help with creating the nets for the real-world application problems. Allow students to use a net cutout to view the three-dimensional object as well as the two-dimensional net.

LITERACY AND LANGUAGE ROUTINES

Clarify, Critique, Correct

Consider using a Clarify, Critique, Correct routine prior to completing the 8.11.2 Guided Instruction. Pose an incorrect statement about the height and the slant height being the same heights. Allow the students to critique the statement, clarify the differences between the heights, and correct the statement.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

This standard focuses on validating student understanding with the use of different representations. By allowing students to find the surface area using photographs of real-life examples, models of solids, and images of nets, they will be using a variety of representations to establish comprehension.

DIFFERENTIATE INSTRUCTION

Provide multiple representations and opportunities to connect the two-dimensional shapes in a net to the three-dimensional figure of a pyramid. Encourage students to think of other real-world examples of pyramids, find online images of different pyramids, or print various nets of different-sized pyramids for students to cut out and fold, and use these nets to demonstrate the difference between slant height and pyramid height. By providing visual, digital, and physical representations of the pyramid and corresponding nets, students connect the two-dimensional and three-dimensional versions and develop strategies to find the surface area.

8.11.1 WARM-UP: WOULD YOU RATHER?

Component Overview: The Warm-Up provides two real-world photographs of two bedrooms and provides the length and the perimeter of each room. Students must decide which room they would rather have and write an explanation of their selection.

TEACHER MOVES

Provide sentence stems to assist students with clearly communicating their explanations in question 2:

- “I would rather have the ____ room because it _____.”
- “I like the ____ about the ____ room because _____.”



8.11.1 Warm-Up: Would You Rather?

1. Which room would you rather have?

	Room 1	Room 2
Length	20 feet	14 feet
Perimeter	58 feet	56 feet

2. Write an explanation of your choice.

Answers will vary.

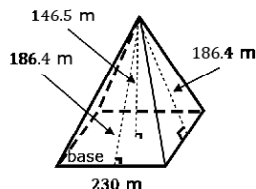
I want the Room 1 because the length is bigger.

I want the Room 2 because I have 56 feet of lights to put around the room which aren't long enough for Room 1.



8.11.2 Guided Instruction: Surface Area of a Rectangular Pyramid

1. Around 2500 B.C. the Egyptians began building pyramids with intricate hallways and layouts to house the tombs of pharaohs and their families. One of the most famous is the Pyramid of Khufu, seen below. The Egyptians designed the pyramid to have a square base with approximate side lengths of 230 meters, a height of 146.5 meters, and a slant height of 186.4 meters.



Looking at the model of the pyramid, why is there a measurement called height and a measurement called slant height?

The height is straight up the middle of the pyramid to the highest point. The slant height is up the side of the pyramid to the top because the side is slanted.

The slant height of the pyramid represents the height of each triangular face.



When finding the surface area of a pyramid, use the **slant height** to find the area of the faces.

8.11.2 GUIDED INSTRUCTION: SURFACE AREA OF A RECTANGULAR PYRAMID

Component Overview: Students are introduced to pyramids using Egyptian pyramids and focus on finding the surface area when given a 3D shape and/or net. Instruction should include the difference between the height and the slant height. Students label the net, find the area of each face, and find the total surface area, then select correct options to complete statements on finding the surface area of pyramids.

QUESTIONING

While looking at the model of the pyramid in question 1, prompt students to consider the key elements distinguishing the height from the slant height by asking:

- Where in the diagram are the height of the pyramid and the slant heights of the faces located?
- Why does the diagram include more than one slant height and only one height?
- What does the square box at the bottom of the height indicate?

DIFFERENTIATE INSTRUCTION

Consider using a hands-on demonstration to help students differentiate between slant height and pyramid height. Use a physical net of a pyramid and ask students to identify and point at the slant height of each triangular face on the physical net and on the drawn net in question 1. Then fold the net together and ask students to identify the pyramid height on the physical net and on the drawn pyramid.

Students may find it easier to understand slant height by comparing it to the stairs needed to walk up the side of the pyramid.

8.11.2 GUIDED INSTRUCTION: SURFACE AREA OF A RECTANGULAR PYRAMID (CONTINUED)

TEACHER MOVES

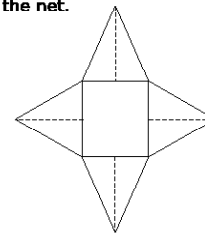
Encourage students to analyze the net provided prior to completing the table. Students may point out that the center shape is a square because all the side lengths are the same. Students may also note that all the faces attached to the square are triangles. Students may also notice that squares are special kinds of rectangles.

Also compare the net of a pyramid to the net of a rectangular prism and look for similarities and differences, like the number and shape of the faces.

DIFFERENTIATE INSTRUCTION

Consider informally observing student work and using observations to determine student groups for a jigsaw activity in 8.11.4 Your Turn!.

2. Using the information about the pyramid, complete the table to determine the surface area of the pyramid. Begin by labeling the sides of the net.



Side	Formula	Area
Rectangular base	$A = l \cdot w$	$A = 230 \cdot 230$ $A = 52,900 \text{ m}^2$
Triangular Face 1	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 230 \cdot 186.4$ $A = 21,436 \text{ m}^2$
Triangular Face 2	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 230 \cdot 186.4$ $A = 21,436 \text{ m}^2$
Triangular Face 3	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 230 \cdot 186.4$ $A = 21,436 \text{ m}^2$
Triangular Face 4	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 230 \cdot 186.4$ $A = 21,436 \text{ m}^2$
		Surface Area: $138,644 \text{ m}^2$

3. Complete the following statement.

To find the surface area of a rectangular pyramid you need

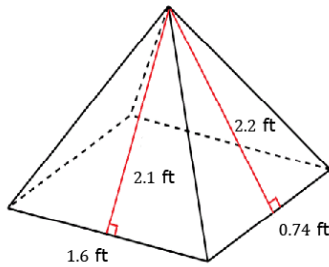
the sum of the ☐ height of the base plus the area of

☐ one of ☐ area the other faces.



8.11.3 Collaboration: Surface Area of a Rectangular Pyramid

Papier-mâché is a type of art that uses paper and glue. Jose is using the art of papier-mâché to create a rectangular pyramid for his art project. To create his project, Jose needs to determine the amount of area that will be covered in newspaper. Below is the drawing Jose created to find the surface area.



1. Work with your partner to find the surface area of the pyramid.

$$A_{\text{rectangular face}} = 1.6 \times 0.74 = 1.184 \text{ ft}^2$$

$$A_{\text{triangular face 1}} = \frac{0.74 \times 2.2}{2} = 0.814 \text{ ft}^2$$

$$A_{\text{triangular face 2}} = \frac{1.6 \times 2.1}{2} = 1.68 \text{ ft}^2$$

$$A_{\text{triangular face 3}} = \frac{0.74 \times 2.2}{2} = 0.814 \text{ ft}^2$$

$$A_{\text{triangular face 4}} = \frac{1.6 \times 2.1}{2} = 1.68 \text{ ft}^2$$

$$\begin{aligned} \text{Total Surface Area} &= 1.184 + 0.814 + 1.68 + 0.814 + 1.68 \\ &= 6.172 \text{ ft}^2 \end{aligned}$$

8.11.3 COLLABORATION: SURFACE AREA OF A RECTANGULAR PYRAMID

Component Overview: Students work with a partner to find the total surface area of a labeled diagram of a real-world example of a rectangular pyramid.

ENRICHMENT

To scaffold students through the process to find the surface area, pose these questions:

- What are the five shapes that make up this pyramid?
- Which of these shapes are congruent?
- What are the dimensions of the base and of each face?
- How can the area of the shapes be found?
- Once the areas are found, how can the total surface area be found?

DIFFERENTIATE INSTRUCTION

Finding the surface area of three-dimensional figures is a process that involves many steps for students to execute. Encourage students to draw labeled pictures as they go through the steps to find the surface area. Also encourage students to use a table, like the table included in the 8.11.2 Guided Instruction, for them to organize their calculations.

QUESTIONING

Consider an interdisciplinary lesson with an art teacher using papier-mâché to conceptualize and represent surface area. Have students create their own version of a net of a pyramid using cardstock paper, then measure the surface area to determine how much newspaper they would need to cover the pyramids with three layers. If time allows, let students create these pyramids and papier-mâché the outside surface area to display in the classroom.

8.11.4 YOUR TURN!

Component Overview: Students find the total surface area of two labeled images and one real-world example of rectangular pyramids. Consider using a jigsaw routine in which students solve one problem and are responsible for sharing their learning with two other students who solved the other two problems.

ENRICHMENT

Elicit the problem-solving process by asking questions to support students as they find the surface area, like:

- What do you need to do first to find the surface area? Where should you start?
- What are the base and the height of the triangles?
- Once the areas of all the faces are found, what operation is used to find the total surface area?
- How can you check if your answer is reasonable?

TEACHER MOVES

Prior to completing question 2, prompt students to notice specific characteristics of the pyramid. Students may recognize that the height of the pyramid is not necessary to calculate the surface area. Students may also see that since two different slant heights are labeled, all the triangular faces are not identical.

TEACHER MOVES

Consider asking students what they know about Paris or the Louvre Museum. Students who enjoy art may know some of the famous art pieces on display at the Louvre. Ask students if they have ever visited an art museum or if they would like to visit Paris one day.

QUESTIONING

For question 3, consider providing the students with a rectangular pyramid with alternate dimensions to the Louvre Museum such that the shape of the base still results in the same area, such as 68 meters by 17 meters.

Allow students to simultaneously practice finding the surface area of the pyramid with a square base and finding the surface area of the same pyramid with a rectangular base.

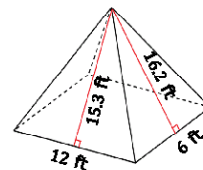
This demonstrates that finding the surface area of all rectangular pyramids is the same, regardless of the shape of the base.



8.11.4 Your Turn!

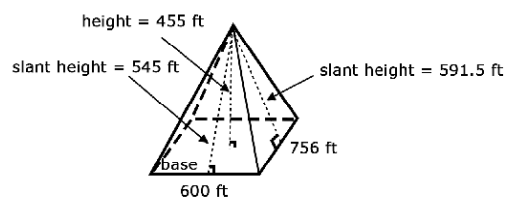
Calculate the surface area for the pyramids.

1. Calculate the surface area of the pyramid.



$$\text{Surface Area} = 352.8 \text{ ft}^2$$

2. Find the surface area of a rectangular pyramid that has a rectangular base with a side length of 600 feet (ft), which is the base of the triangle with a slant height of 545 ft, and a side length of 756 ft, which is the base of the triangle with a slant height of 591.5 ft.



$$\text{Surface Area} = 1,227,774 \text{ ft}^2$$

3. At the entrance to the Louvre Museum in Paris, France, is an architectural pyramid that was completed in 1989. The pyramid has a square base with side lengths of 34 meters, and the pyramid stands at a height of 21.6 meters and a slant height of 27.5 meters. Using this information, calculate the surface area of the rectangular pyramid.

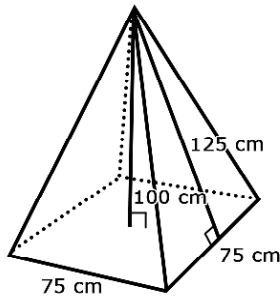


$$\text{Surface Area} = 3,026 \text{ m}^2$$



8.11.5 Wrap-Up: Ticket Out the Door

- Find the surface area of a rectangular pyramid that has a square base with side lengths 75 centimeters (cm), a height of 100 cm, and a slant height of 125 cm.



Surface Area = $24,375 \text{ cm}^2$

8.11.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Students individually find the total surface area of a labeled diagram of a square pyramid. Consider using student responses as a data point on student learning and mastery of the lesson's learning target.

DIFFERENTIATE INSTRUCTION

Prompt students to draw a net of the pyramid to visualize the five faces. Encourage them to label all the dimensions and to label the area of each shape on their net once they calculate them. Allowing the students to create an alternate visual will assist them with finding the surface area of the solid as well as the net of the same pyramid.